

ESPIRITU SANTO-PEKOA, Vanuatu

WMO: 915540

Lat: 15.501S

Lon: 167.223E

Elev: 56

StdP: 100.65

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.2	20.4	15.2	10.9	23.1	16.3	11.7	23.5	8.9	25.6	8.3	25.4	1.3	300	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.2	31.1	26.4	30.7	26.2	30.2	26.0	27.2	29.9	27.0	29.7	26.7	29.4	3.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.2	29.0	26.2	21.7	28.7	25.9	21.4	28.6	86.9	30.2	85.4	29.8	84.2	29.5	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 8.5	7.7	6.9		17.5	32.6	1.3	1.6	16.6	33.8	15.9	34.7	15.2	35.6	14.2		
(5)				16.1	27.9	1.1	0.6	15.3	28.3	14.7	28.6	14.1	29.0	13.3	29.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	26.9	27.0	26.9	26.6	25.8	25.1	24.8	24.6	25.1	25.6	26.2	26.7	(6)	
(7)		DBStd	1.34	0.89	0.83	0.74	0.78	1.12	1.20	1.17	1.26	1.10	1.15	1.12	0.93	(7)	
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
(10)		CDD10.0	5817	526	476	523	497	491	454	457	452	454	484	486	518	(10)	
(11)		CDD18.3	2776	267	242	264	247	232	204	199	194	204	226	236	259	(11)	
(12)		CDH23.3	23413	2641	2412	2541	2294	1952	1428	1185	1097	1372	1797	2156	2538	(12)	
(13)		CDH26.7	4828	763	713	688	541	291	129	78	84	130	267	438	706	(13)	
(14)	Wind	WSAvg	3.1	2.5	2.4	2.3	2.2	3.1	3.5	3.7	3.9	4.0	3.9	3.4	2.8	(14)	
(15)	Precipitation	PrecAvg	2315	275	264	327	254	185	162	108	109	107	143	201	180	(15)	
(16)		PrecMax	3355	532	675	886	876	359	503	392	375	314	386	510	343	(16)	
(17)		PrecMin	1138	0	0	121	78	16	2	11	15	2	0	41	24	(17)	
(18)		PrecStd	538	145	150	159	157	113	120	89	91	83	102	112	90	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.7	31.8	31.2	31.0	30.0	29.5	29.1	29.2	29.5	30.0	30.7	31.2	(19)	
MCWB			26.6	26.5	26.5	26.8	26.0	25.6	24.9	24.9	24.7	25.1	26.0	26.3	(20)		
2%		DB	31.0	31.1	30.6	30.2	29.3	28.7	28.1	28.3	28.7	29.2	30.1	30.7	(21)		
		MCWB	26.4	26.3	26.4	26.3	25.5	25.2	24.4	24.4	24.5	24.6	25.6	26.1	(22)		
5%		DB	30.3	30.5	30.2	29.8	28.9	28.1	27.5	27.7	28.0	28.8	29.5	30.1	(23)		
		MCWB	26.0	26.2	26.2	26.0	25.2	24.7	24.0	23.9	24.0	24.4	25.2	25.8	(24)		
10%		DB	29.9	30.0	29.8	29.3	28.2	27.5	26.9	27.0	27.3	28.2	29.0	29.7	(25)		
		MCWB	25.8	26.1	26.0	25.8	24.7	24.3	23.6	23.4	23.6	24.1	24.8	25.6	(26)		
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.7	27.7	27.4	27.5	26.8	26.6	26.0	25.8	26.2	26.5	26.8	27.2	(27)
MCDWB				30.5	30.2	29.8	29.9	29.1	28.3	27.7	28.0	27.6	28.3	29.5	30.1	(28)	
2%	WB		27.1	27.2	27.0	26.9	26.3	26.0	25.3	25.1	25.4	25.8	26.2	26.8	(29)		
	MCDWB		29.9	29.9	29.7	29.6	28.5	27.9	27.1	27.5	27.2	27.9	28.9	29.7	(30)		
5%	WB		26.7	26.8	26.7	26.6	25.9	25.5	24.8	24.7	25.0	25.3	26.0	26.4	(31)		
	MCDWB		29.4	29.6	29.4	29.1	28.0	27.3	26.7	26.9	27.0	27.6	28.6	29.3	(32)		
10%	WB		26.3	26.5	26.3	26.2	25.5	25.0	24.3	24.2	24.6	24.8	25.5	26.1	(33)		
	MCDWB		29.0	29.2	29.0	28.6	27.5	26.9	26.2	26.2	26.6	27.2	28.1	28.9	(34)		
(35)	Mean Daily Temperature Range		5% DB	MDBR	6.0	6.2	5.8	5.4	4.9	4.5	4.3	4.6	5.1	5.4	6.1	(35)	
MCDBR				6.8	7.1	6.6	6.4	5.7	5.4	5.4	5.5	5.6	6.0	6.3	6.7	(36)	
MCWBR		3.4		3.4	3.2	3.2	3.0	2.9	2.9	3.0	3.0	2.9	3.0	3.3	(37)		
5% WB		MCDWB	6.4	6.4	6.2	6.2	5.2	4.9	4.9	5.1	4.9	5.3	5.4	6.4	(38)		
		MCWBR	3.5	3.4	3.3	3.4	3.0	3.1	3.1	3.1	2.9	2.9	3.0	3.4	(39)		
		(40)	Clear-Sky Solar Irradiance	taub	0.403	0.395	0.392	0.387	0.392	0.396	0.400	0.399	0.405	0.410	0.408	0.401	(40)
taud	2.545	2.576		2.590	2.607	2.568	2.553	2.516	2.515	2.489	2.490	2.514	2.537	(41)			
Ebn at Noon	941	940		923	894	854	833	837	870	897	915	931	943	(42)			
Edh at Noon	110	106		102	94	92	91	96	102	111	114	113	111	(43)			
(44)	All-Sky Solar Radiation	RadAvg	6.11	5.82	5.36	4.71	4.09	3.70	3.91	4.51	5.26	5.69	5.91	6.33	(44)		
RadStd		0.58	0.50	0.49	0.41	0.40	0.24	0.23	0.32	0.46	0.52	0.56	0.45	(45)			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page